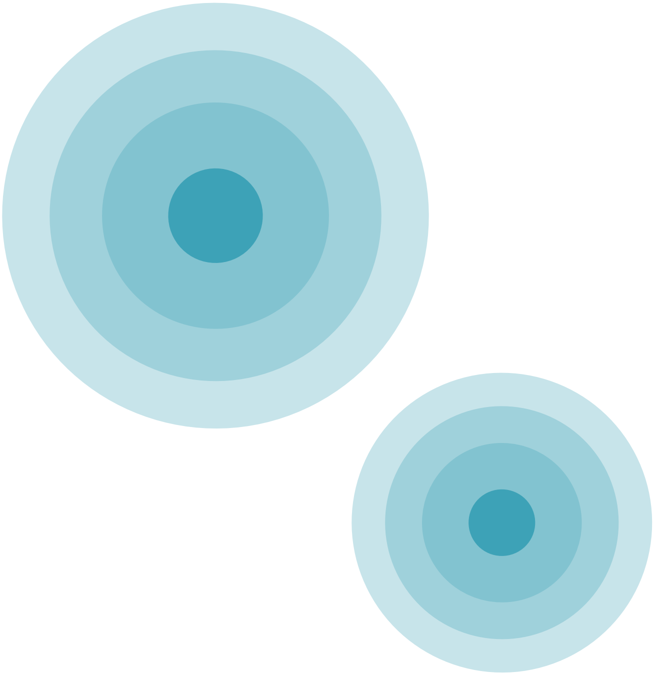
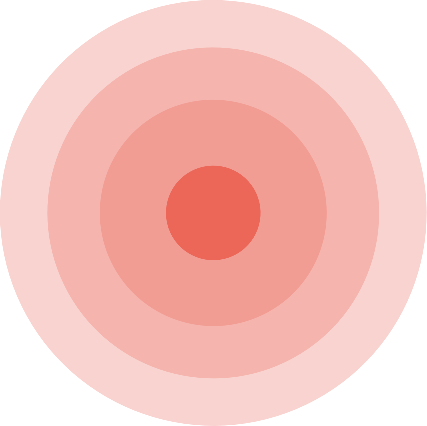
pushforward





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cocap	teal	salmon	citrus	beige	copper

Push-forwards

Measurable spaces X, Y. Measurable function $f : X \to Y$.

The **pushforward** of $\alpha \in \mathscr{M}(X)$ by f is the measure given by

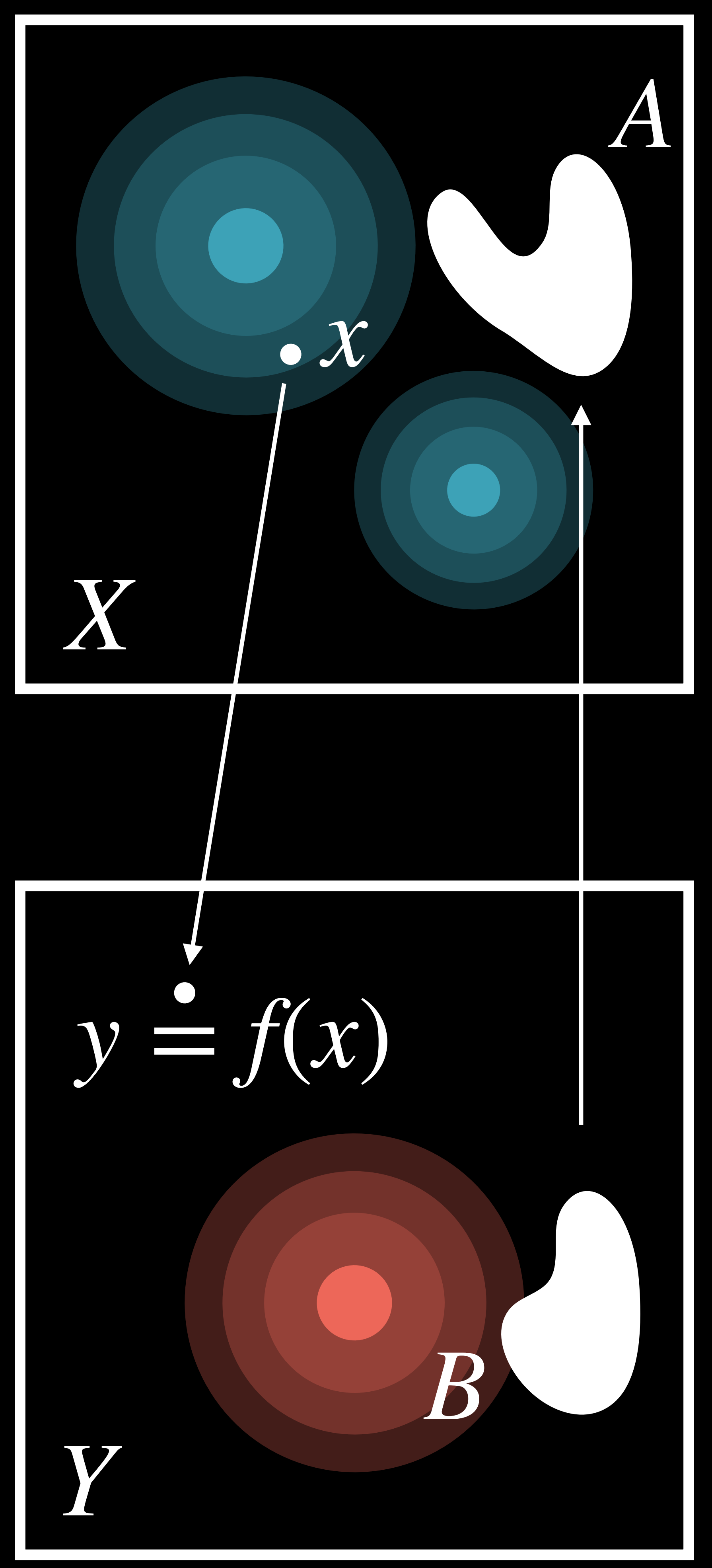
$$f_\sharp(\alpha)(B) := \alpha(f^{-1}(B)), \quad B \subset Y$$

Alternative characterization (change of variable formula). For any measurable function g:

$$\int_Y g(y)\,\mathrm{d}[f_\sharp(\alpha)(y)] = \int_X g(f(x))\mathrm{d}\alpha(x)$$

For measures w/ smooth densities $\mathrm{d}\alpha = \rho(x)\mathrm{d}x$, $\mathrm{d}\beta = \xi(x)\mathrm{d}x$:

$$f_\sharp\alpha = \beta \iff \rho(f(x))\,|\det(Df(x))| = \xi(x)$$



Pop-Quiz

What is the push-forward of a discrete probability measure $\alpha = \sum_{1}^{n} \mathbf{a}_i \delta_{x_i}$?